Three-Phase controlled Rectifiers

Three-Phase Half-Wave Controlled Rectifier

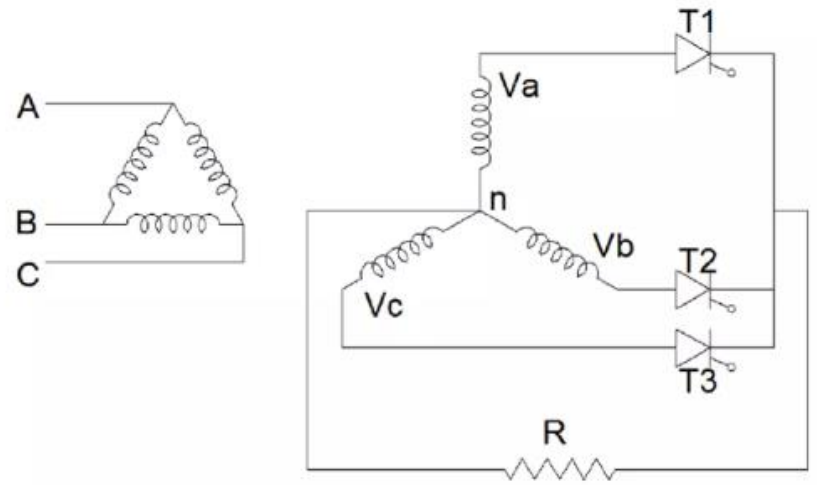
- A three-phase half-wave controlled rectifier is an electrical circuit used for converting alternating current (AC) into direct current (DC) by controlling the flow of current through a load. It is called a half-wave rectifier because it rectifies only one half of the input waveform.
- Each phase of the AC source is connected to a thyristor, forming a thyristor bridge. The diodes allow current flow in one direction and block it in the other direction. When the input voltage is positive in one phase, the corresponding thyristor conducts provided that gate trigger is given, allowing current to flow through the load connected to the output side of the bridge.

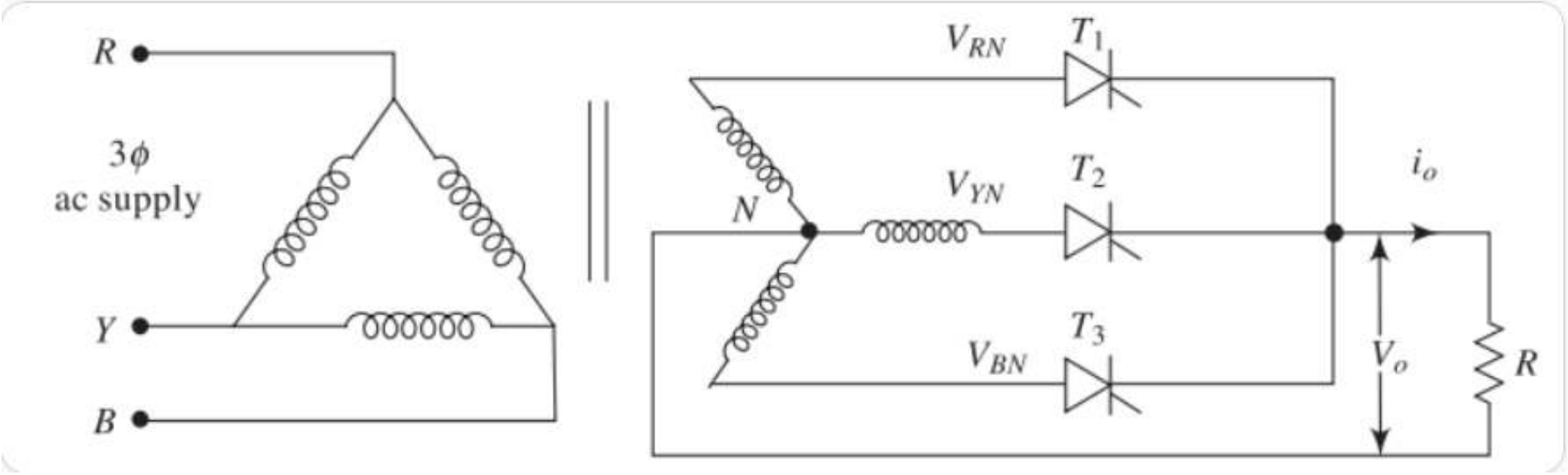
In a controlled rectifier, the current flow through the thyristor (also known as a silicon-controlled rectifier or SCR). The thyristor acts as a switch that can be turned on or off depending on the control signal applied to its gate.
o control the rectification process, the thyristors are triggered at specific points in the input waveform. Triggering is achieved by applying a pulse signal to the gate of the thyristor at the desired firing angle. The firing angle determines the portion of the input waveform during which the thyristor conducts current.
Once triggered, the thyristor turns on and allows current to flow through the load until the end of the half-cycle or until the thyristor is turned off.
When the input waveform crosses zero or becomes negative, the natural commutation occurs, turning off the conducting thyristor. This happens because the forward voltage across the thyristor drops below its holding voltage.

The load connected to the output of the rectifier receives a pulsating DC voltage, which consists of positive half-cycles. The output voltage waveform is the result of rectifying only one half of the input waveform.

By controlling the firing angle of the thyristors, the average output voltage can be varied, enabling the regulation of power delivered to the load.
It's important to note that the three-phase half-wave controlled rectifier is a simplified configuration and may require additional components, such as filters and smoothing capacitors, to reduce ripple and provide a smoother DC output. Overall, the three-phase half-wave controlled rectifier provides a means to convert three-phase AC power to pulsating DC power by controlling the firing of thyristors in each phase.

Three-Phase Half-Wave Controlled Rectifier - R Load





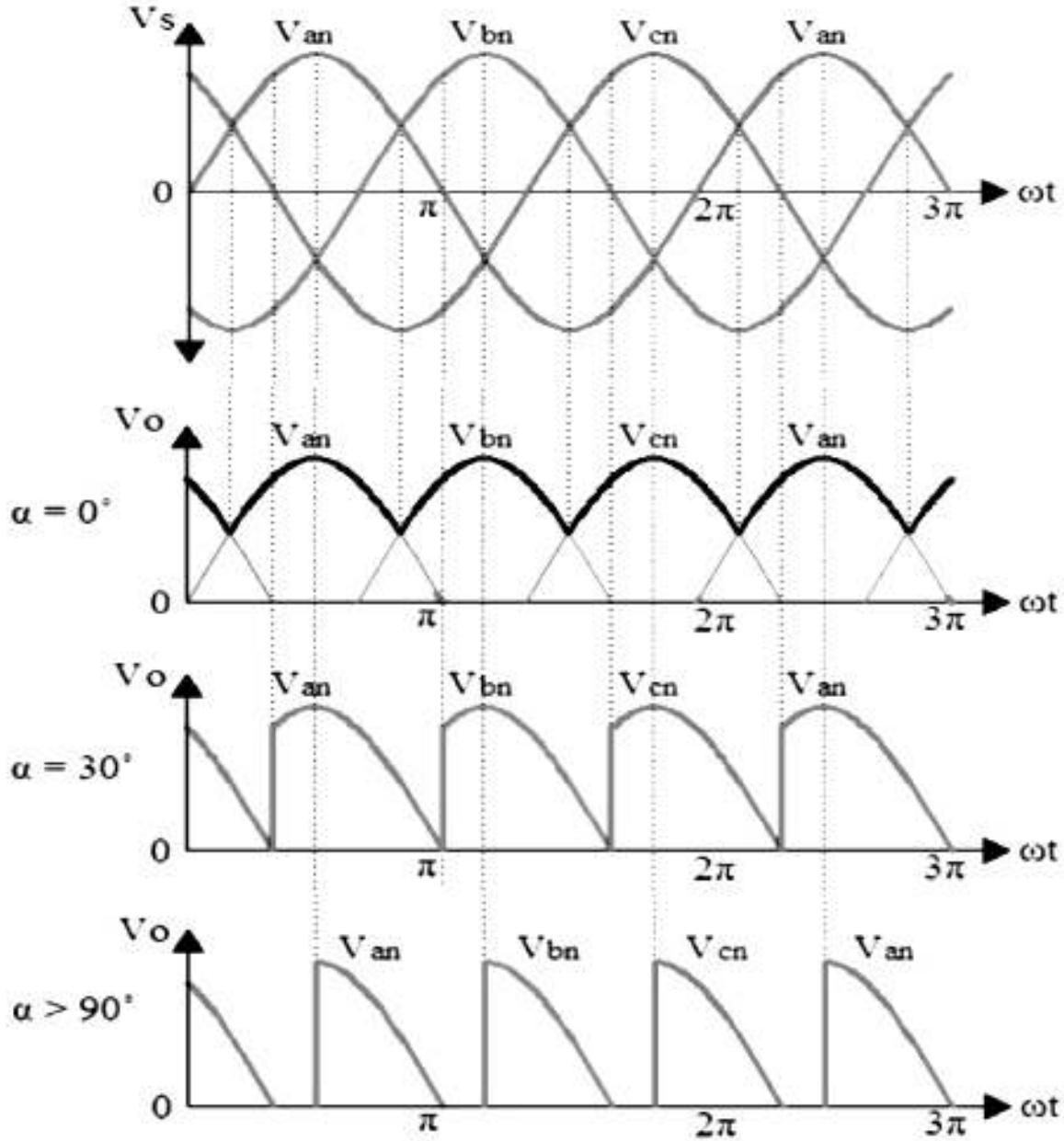
Conducting period

Device	Starting Period	Ending period
T_1	30°+α	150°+α
T_2	150°+α	270°+α
T_3	270°+α	390°+α

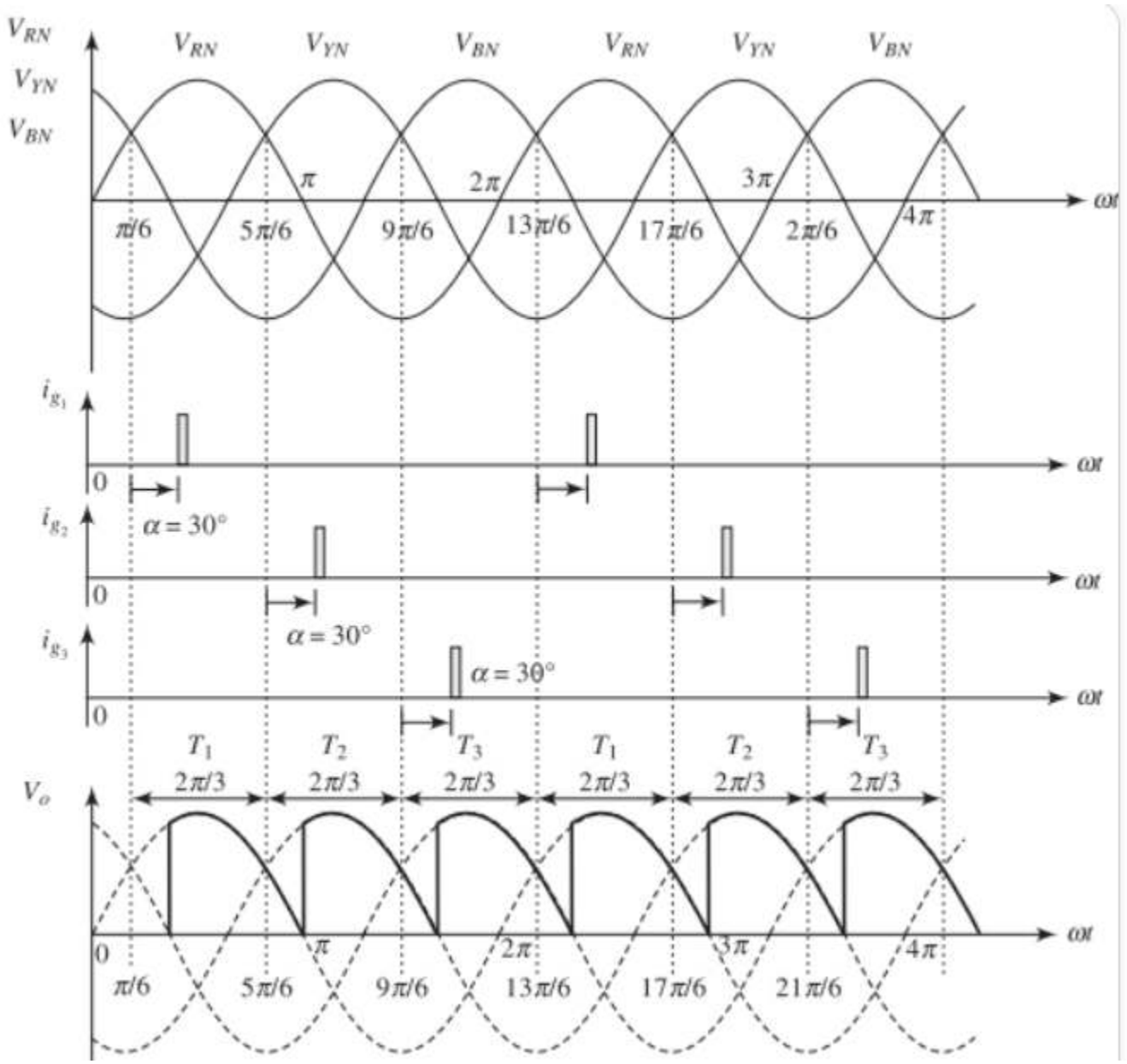
Example $\alpha = 15°$

Device	Starting Period	Ending period
T_1	30°+α = 45°	150°+α = 165°
T_2	150°+α = 165°	270°+α = 285°
T_3	270°+α = 285°	390°+α = 405° (or 45°)

The circuit consist of a delta star transformer and 3 thyristors T_1, T_2, T_3 which are connected on the secondary star connected winding and a resistive load. When a V_a is positive, T_1 becomes forward biased and conducts. During the negative cycle of V_a, T_1 turns off. Similarly T_2 and T_3 conducts only during the positive cycles of V_b and V_c respectively. The average output voltage can be varied by varying the firing angles of the thyristors. The waveforms show the output voltage for various firing angles. In the waveform, V_a is denoted as V_{an}, V_b as V_{bn} and V_c as V_{cn}.

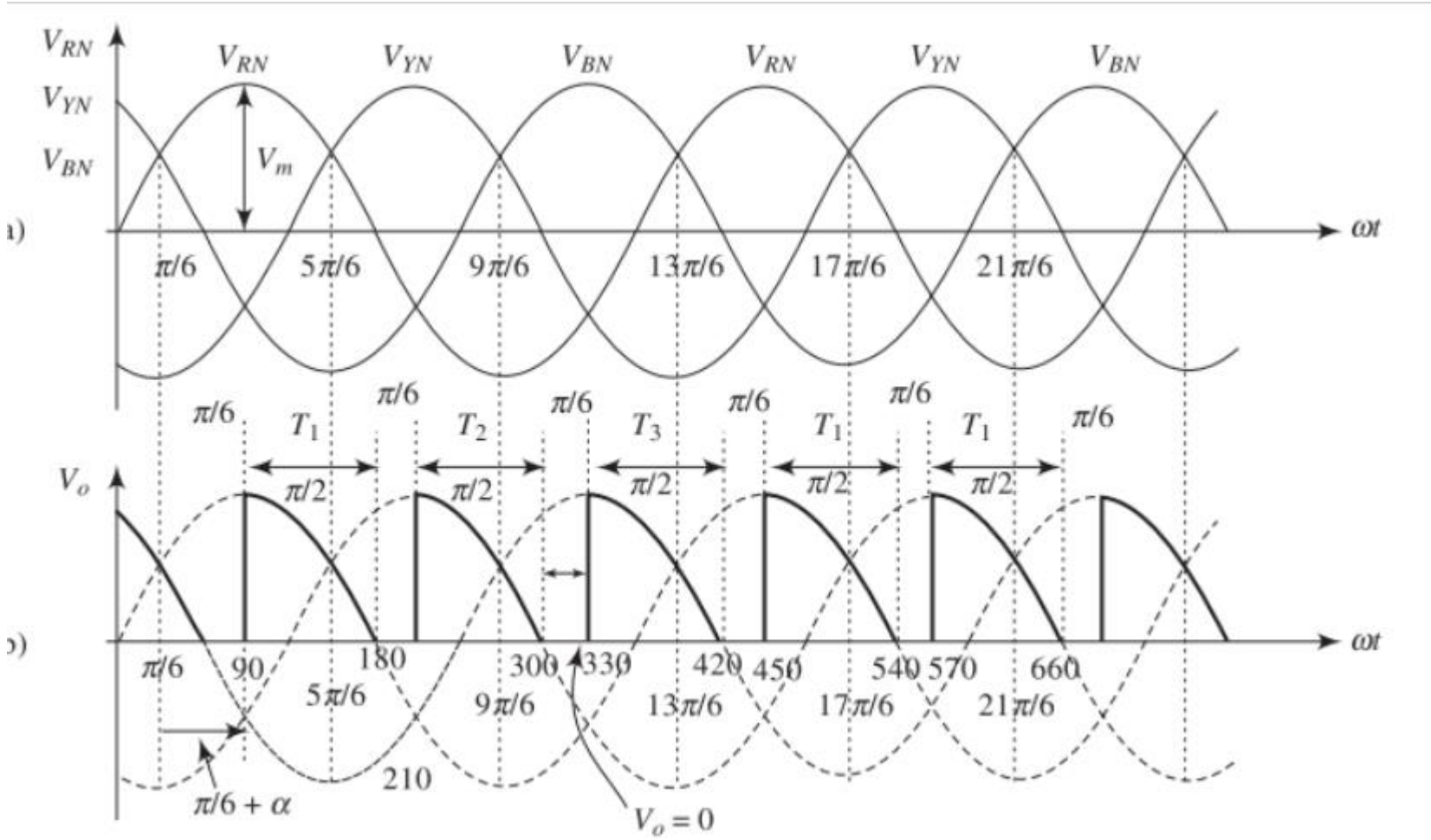

Vs
Van
Vbn
Vcn
Van
0
π
2π
3π
ωt
Vo
α = 0°
Van
Vbn
Vcn
Van
0
π
2π
3π
ωt
Vo
α = 30°
Van
Vbn
Vcn
Van
0
π
2π
3π
ωt
Vo
α > 90°
Van
Vbn
Vcn
Van
0
π
2π
3π
ωt

For $\alpha = 30°$


V_{RN}
V_{YN}
V_{BN}
V_{RN} V_{YN} V_{BN} V_{RN} V_{YN} V_{BN}
π 2π 3π 4π
$\pi/6$ $5\pi/6$ $9\pi/6$ $13\pi/6$ $17\pi/6$ $21\pi/6$
ωt
i_{g_1}
i_{g_2}
i_{g_3}
0
$\alpha = 30°$
T_1 T_2 T_3 T_1 T_2 T_3
$2\pi/3$
V_o

For $\alpha > 30°$

$\alpha = 60°$



The expression for average and rms of output voltage and current are:

For $0 < \alpha \leq 30°$ (Output voltage waveform is continuous)

$$V_{av} = \frac{3}{2\pi}\int_{30+\alpha}^{150+\alpha} V_o \, d(\omega t)$$

$$= \frac{3V_m}{2\pi}\int_{30+\alpha}^{150+\alpha} \sin \omega t \, d(\omega t)$$

$$= \frac{3Vm}{2\pi}\left[-\cos \omega t\right]$$

$$= \frac{3Vm}{2\pi}\left[-(\cos(150+\alpha) - \cos(30+\alpha))\right]$$

$$= 3\frac{\sqrt{3}}{2}\frac{\mathrm{Vm}}{\pi}\cos \alpha$$

$$V_o = \frac{1}{\frac{2\pi}{3}} \int_{\alpha+\frac{\pi}{6}}^{\alpha+\frac{5\pi}{6}} V_{mp}\, sinwt\, dwt = \frac{3\sqrt{3}}{2\pi} V_{mp} \cos\alpha = \frac{3}{2\pi} V_{ml} \cos\alpha.$$

$$I_o = \frac{V_o}{R} = \frac{3}{2\pi R} V_{ml} \cos\alpha.$$

$$V_{o\,(rms)} = \left[\frac{1}{\frac{2\pi}{3}} \int_{\alpha+\frac{\pi}{6}}^{\alpha+\frac{5\pi}{6}} V_{mp}^2\, sin^2 wt\, dwt\right]^{1/2}$$

$$V_{o\,(rms)} = \sqrt{3} V_{mp} \left[\frac{1}{6} + \frac{\sqrt{3}}{8\pi} cos2\alpha\right]^{1/2} = V_{ml} \left[\frac{1}{6} + \frac{\sqrt{3}}{8\pi} cos2\alpha\right]^{1/2}$$

$$V_{\text{rms}} = V_m \sqrt{\frac{3}{2\pi}\left[\frac{\pi}{3} + \frac{\sqrt{3}}{4}\cos 2\alpha\right]}$$

$$I_{o\,rms} = \frac{V_{o\,rms}}{R} = \frac{\sqrt{3} V_{mp}}{R} \left[\frac{1}{6} + \frac{\sqrt{3}}{8\pi} cos2\alpha\right]^{1/2}$$

For 30° < α < 150° (Output voltage waveform is discontinuous)

$$V_{av} = \frac{3}{2\pi}\int_{30+\alpha}^{\pi} V_o \, d(\omega t)$$

$$= \frac{3V_m}{2\pi}\int_{30+\alpha}^{\pi} \sin \omega t \, d(\omega t)$$

$$= \frac{3Vm}{2\pi}\left[-\cos \omega t\right]$$

$$= \frac{3Vm}{2\pi}\left[-(\cos \pi - \cos(30+\alpha))\right]$$

$$= \frac{3}{2}\frac{\mathrm{Vm}}{\pi}\left[1+\cos(30+\alpha)\right]$$

$$V_o = \frac{1}{\frac{2\pi}{3}} \int_{\alpha+\frac{\pi}{6}}^{\pi} V_{mp}\, sinwt\, dwt = \frac{3}{2\pi} V_{mp} \left[1 + cos(\alpha + 30^\circ)\right]$$

$$I_o = \frac{V_o}{R} = \frac{3}{2\pi R} V_{mp} \left[1 + cos(\alpha + 30^\circ)\right]$$

$$V_{o\,(rms)} = \left[\frac{1}{\frac{2\pi}{3}} \int_{\alpha+\frac{\pi}{6}}^{\pi} V_{mp}^2 \, sin^2 wt \, dwt \right]^{1/2}$$

$$V_{o\,(rms)} = \frac{\sqrt{3} V_{mp}}{2\sqrt{\pi}} \left[\left(\frac{5\pi}{6} - \alpha \right) + \frac{1}{2} sin(2\alpha + \frac{\pi}{3}) \right]^{1/2}$$

$$V_{\text{rms}} = V_m \sqrt{\frac{3}{2\pi} \left[\frac{5\pi}{6} - \alpha + \frac{1}{2} \sin(30^\circ + 2\alpha) \right]}$$

$$I_{o\,(rms)} = \frac{V_{o(rms)}}{R} = \frac{\sqrt{3} V_{mp}}{2\sqrt{\pi} R} \left[\left(\frac{5\pi}{6} - \alpha \right) + \frac{1}{2} sin(2\alpha + \frac{\pi}{3}) \right]^{1/2}$$

Performance Parameters

Form Factor:

$$FF = \frac{V_{\text{rms}}}{V_{\text{dc}}}$$

Ripple Factor:

$$RF = \sqrt{FF^2 - 1} = \sqrt{\left(\frac{V_{\text{rms}}}{V_{\text{dc}}}\right)^2 - 1}$$

DC Output Power:

$$P_{\text{dc}} = V_{\text{dc}} \cdot I_{\text{dc}} = \frac{V_{\text{dc}}^2}{R}$$

Rectification Efficiency:

$$\eta = \frac{P_{dc}}{P_{ac}} \times 100\%$$

For continuous conduction mode with resistive load:

Average Thyristor Current:

$$I_{T(avg)} = \frac{I_{dc}}{3}$$

rms Thyristor Current:

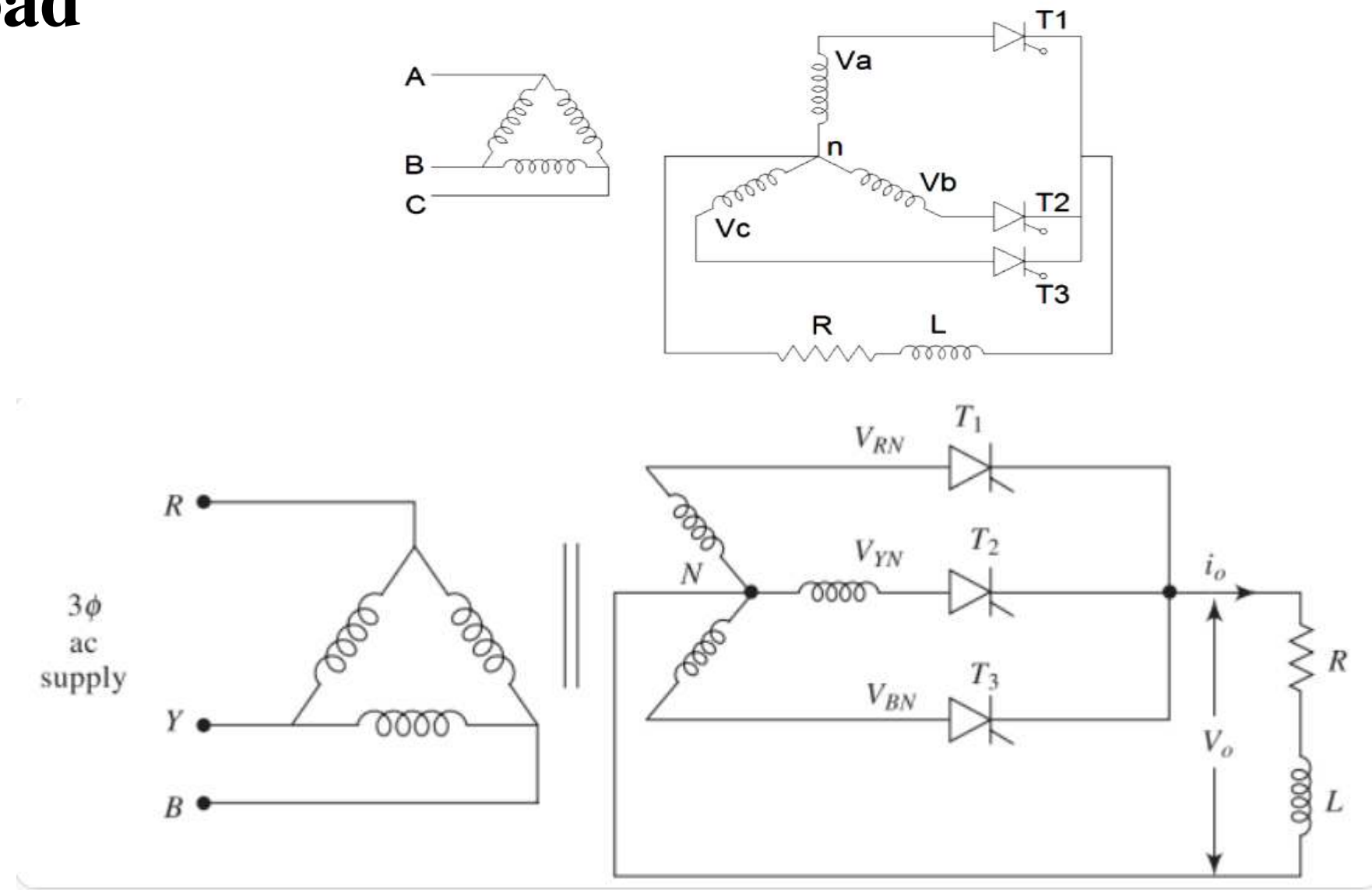
$$I_{T(rms)} = \frac{I_{dc}}{\sqrt{3}}$$

Peak Thyristor Current:

$$I_{T(\text{peak})} = \frac{V_m}{R}$$

Conduction Period: $120°$ or $\frac{2\pi}{3}$ rad per thyristor

Three-Phase Half-Wave Controlled Rectifier - RL Load



The circuit consist of a delta transformer and 3 thyristors T_1, T_2, T_3 which are connected on the secondary star connected winding and a RL load. When V_a is positive, T_1 becomes forward biased and conducts. During the negative cycle of V_a, the current through T_1 is not zero due to inductor present in the load. So T_1 will remian ON during the negative cycle of V_a. When V_b is positive, T_2 is triggered and the load current gets transferred from T_1 toT_2. At this instant, T_1 turns OFF. During the negative cycle of V_b, the current through T_2 is not zero due to inductor present in the load. So T_2 will remain ON during the negative cycle of V_b. When T_3 is triggered during positive cycle of V_c, the load current is transferred from T_2 to T_3. At this instant, T_2 turns OFF. Similarly, T_3 conducts during the negative cycle of V_c and turns OFF when T_1 is triggered. The average output voltage can be varied by varying the firing angles of the thyristors. The waveforms shows the output voltage for various firing angles. In the waveform, V_a is denoted as V_{an}, V_b as V_{bn}, V_c as V_{cn}.

Operating Ranges with RL Load

Range 1: $0 < \alpha < 30°$

- Continuous conduction naturally maintained
- Similar to resistive load behavior

Range 2: $30° < \alpha < 90°$ **(Rectifier Mode)**

Output has negative portions, but $V_{dc} > 0$

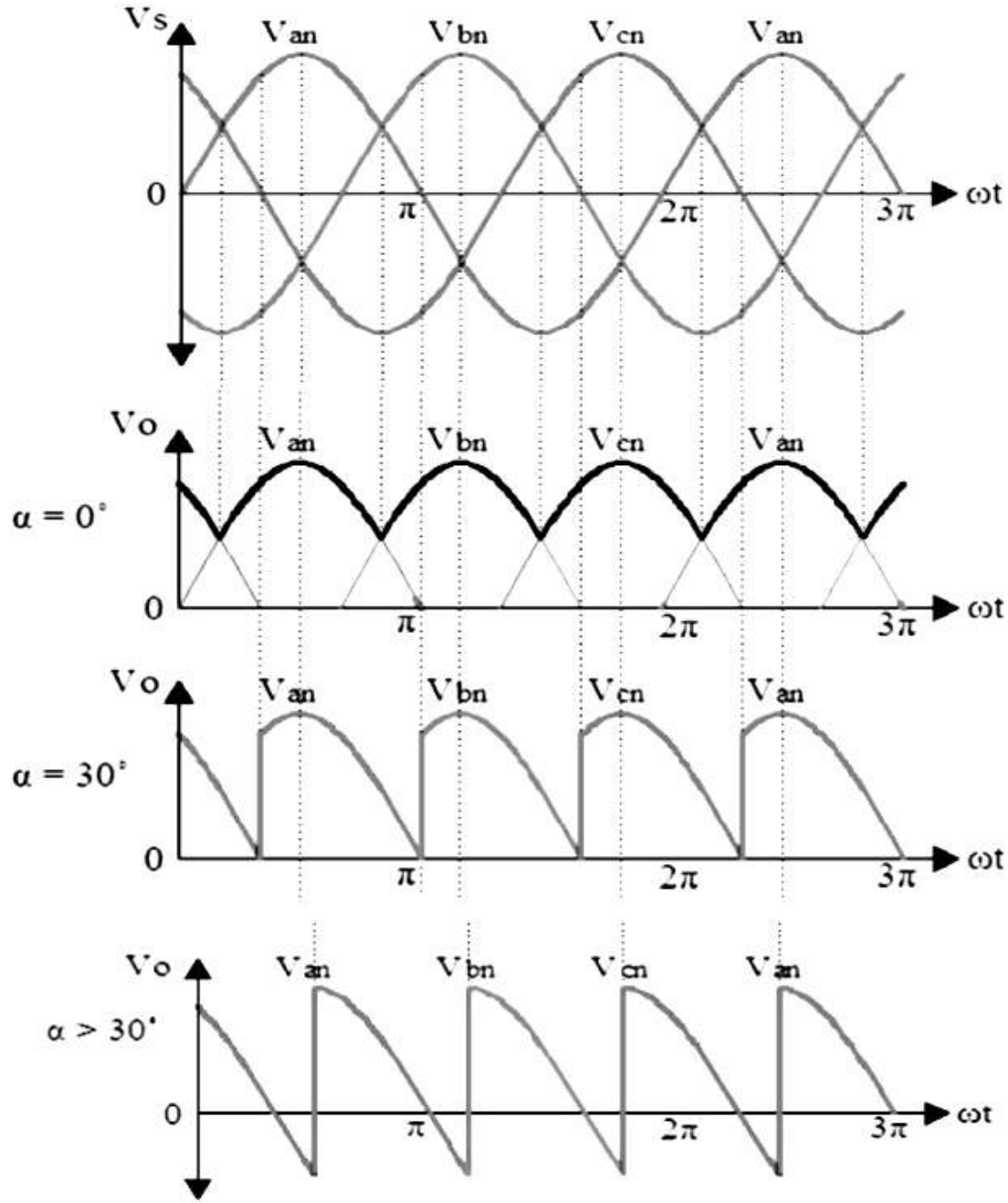
Inductance maintains current continuity

Range 3: $90° < \alpha < 150°$ **(Inverter Mode)**

Average voltage negative: $V_{dc} < 0$

Power flows from DC to AC supply

Requires DC voltage source (e.g., back EMF)


Vs
Van
Vbn
Vcn
Van
0
π
2π
3π
ωt
Vo
α = 0°
Vo
α = 30°
Vo
α > 30°

Waveforms for RL Load: $\alpha = 60°$

Each thyristor: conduction

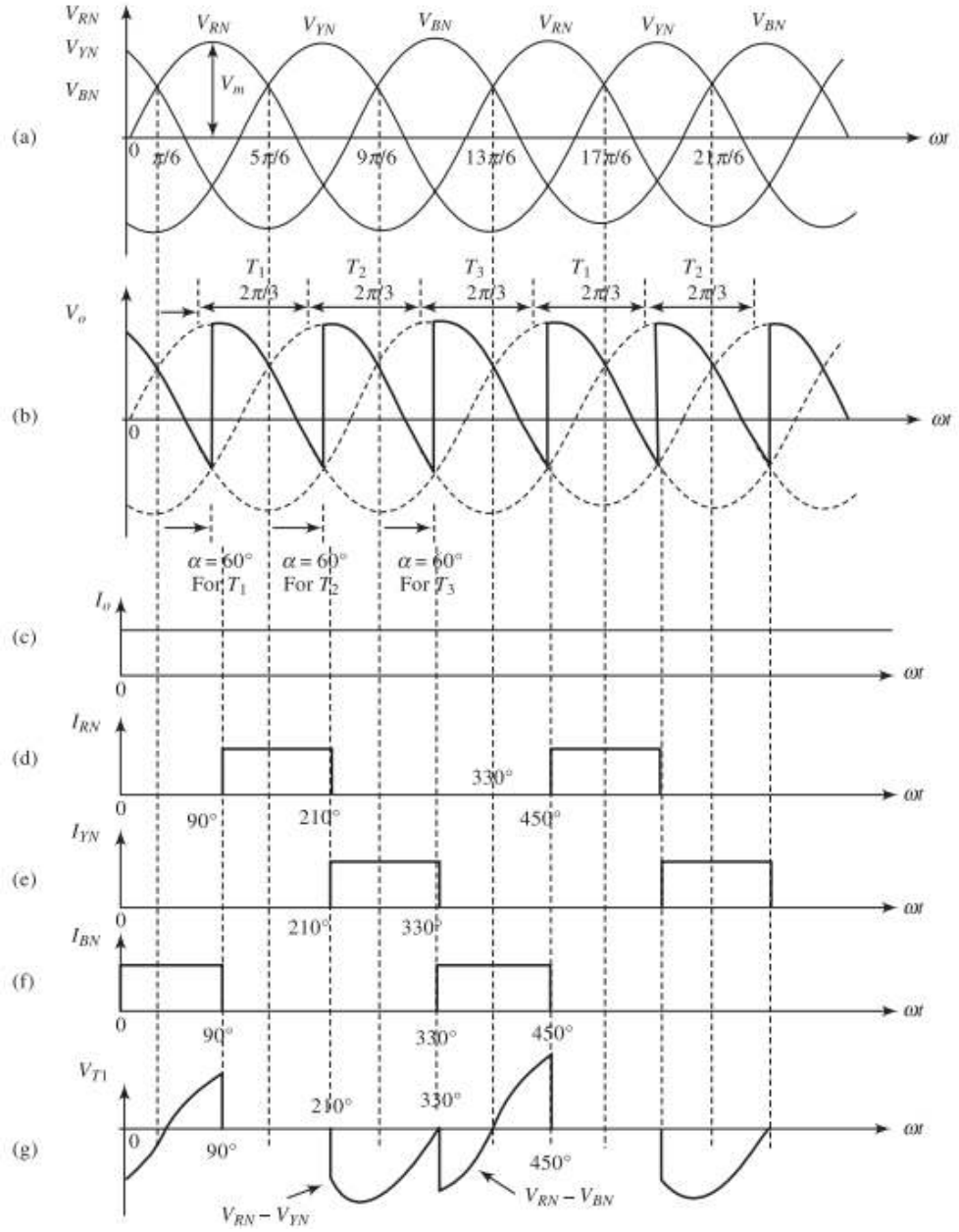
Output voltage contains negative portions

Current remains continuous

Average voltage:

$$V_{dc} = \frac{3\sqrt{3}}{2\pi} V_m \cos\alpha = 0.827\, V_m \cos\alpha$$

$$V_{dc} = 0.414\, V_m \quad \text{(positive)}$$



Rectifier Mode: $\alpha < 90°$

Characteristics:

- Average output voltage positive:
- Power flows from AC supply to DC load
- Energy delivered to the load

Average Output Voltage:

$$V_{dc} = \frac{3\sqrt{3}}{2\pi} V_m \cos\alpha = 0.827\, V_m \cos\alpha$$

Boundary Condition at :$\alpha = 90°$

$$V_{dc} = 0.827\, V_m \cos 90° = 0$$

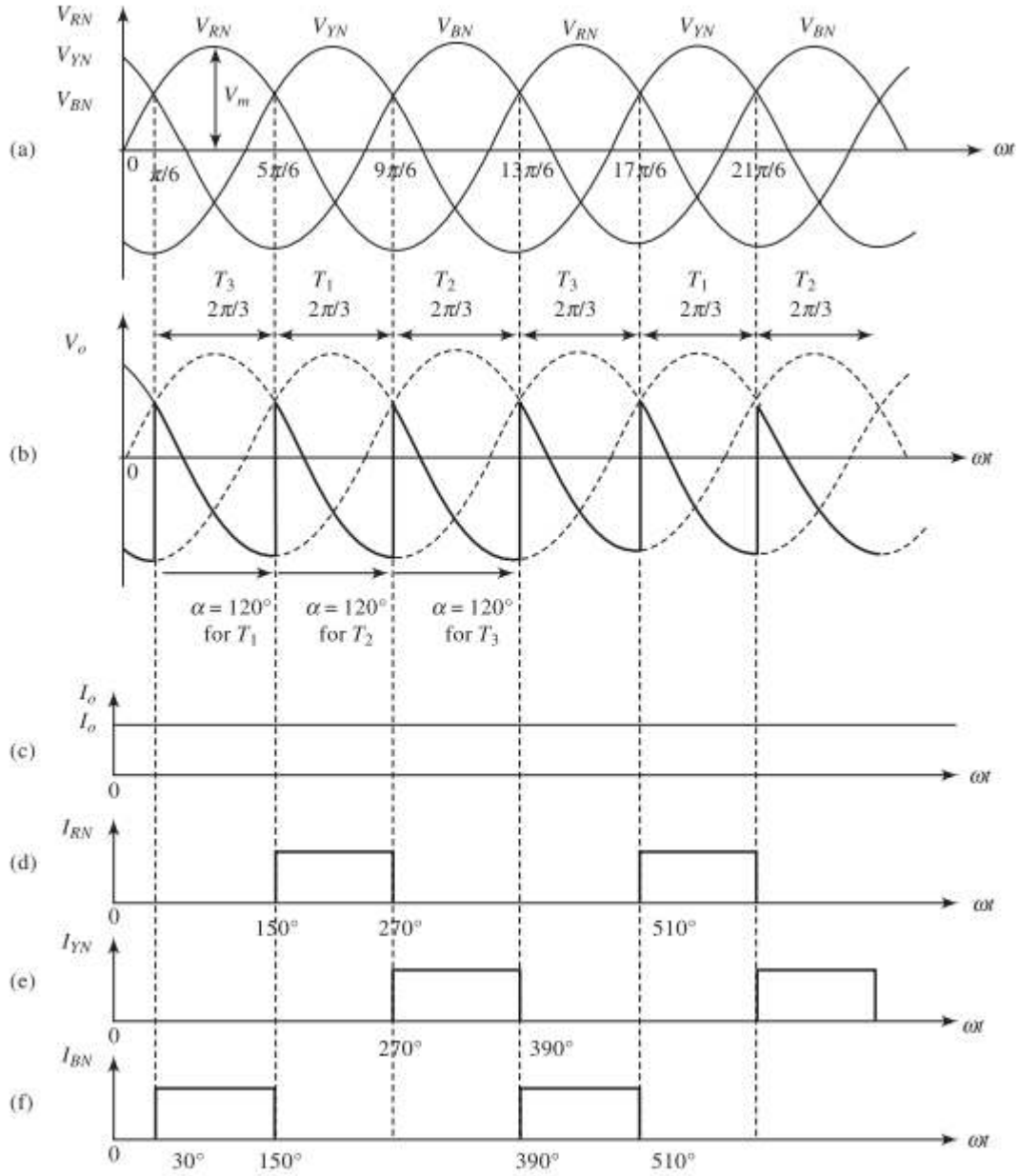
Inverter Mode: $90° < \alpha < 150°$

Example Inverter Mode: $\alpha = 120°$

Characteristics:

- Average voltage negative: $V_{dc} < 0$
- Power flows DC to AC (regeneration)
- Requires DC voltage source
- Used for regenerative braking

$$V_{\text{dc}} = 0.827\, V_m \cos\alpha < 0 \text{ for } \alpha > 90°$$



Voltage and Current Relations with RL Load:

For highly inductive load (continuous current):

Average Output Voltage:

$$V_{\text{dc}} = 0.827\, V_m \cos\alpha$$

Average Load Current (with back EMF):

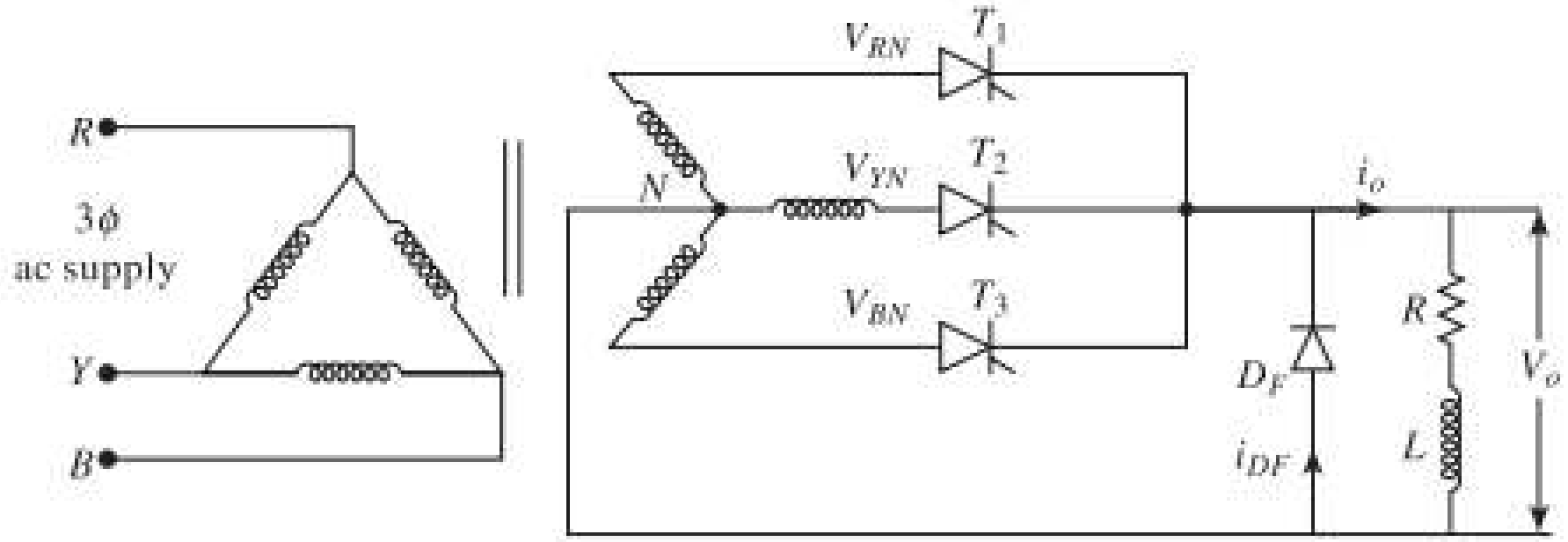
$$I_{\text{dc}} = \frac{V_{\text{dc}} - E}{R}$$

For Pure Inductive Load ($R = 0$)

- Voltage equation: $V_{dc} = E$
- Required firing angle: $\alpha = \cos^{-1}\left(\frac{E}{0.827\ V_m}\right)$
- Conduction: $120°$ per thyristor when $\alpha \leq 150°$

Free-Wheeling Diode Configuration

- **Purpose of FWD:**
- Prevents negative output voltage
- Alternative path for inductive current
- Improves average voltage
- Reduces PIV on thyristors
- Improves load current waveform



Operation with Free-Wheeling Diode

Operation ($\alpha > 30°$):

- FWD conducts when output tends negative
- Thyristor conduction: $(150° - \alpha)$
- FWD conduction: $(\alpha - 30°)$ per cycle
- Output clamped at zero
- Current freewheels through FWD

Output Voltage with Free-Wheeling Diode:

Average DC Voltage (with FWD)

$$V_{\text{dc}} = \frac{3\sqrt{3}}{2\pi} V_m \left[1 + \cos(30^\circ + \alpha)\right]$$

Note: Identical to discontinuous mode with resistive load

Average Thyristor Current:

$$I_{T(\text{avg})} = I_{\text{dc}} \left[\frac{5\pi/6 - \alpha}{2\pi}\right]$$

Average FWD Current:

$$I_{\text{FWD(avg)}} = I_{\text{dc}} \left[\frac{\alpha - \pi/6}{2\pi}\right]$$

Output with FWD: $\alpha = 60^\circ$

